

CareerCode Syllabus

Archer Infotech, Kothrud Pune | Classroom & live-online batches | Updated September 2026

Institute: Archer Infotech, Kothrud, Pune - IT training since 2009

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Programme Overview

Semester-by-Semester Programming and Career Training for Engineering Students — Archer Infotech, Pune

Your engineering degree will give you a certificate. CareerCode will give you a career. College teaches you concepts. CareerCode teaches you how to apply them.

CareerCode is Archer Infotech's flagship ongoing training program for engineering, BCA, and BSc Computer Science students. It runs alongside your college education — semester by semester — adding real technical skills, internship readiness, and career tools to everything your degree provides. From your first year to your final placement drive, CareerCode is your parallel career engine.

Programme Details

- Designed For: Engineering, BCA, and BSc CS students — all years
- Duration: Ongoing — semester by semester throughout your degree
- Pace: 1 to 2 technologies per semester
- Tracks: Frontend / Backend / Full Stack / Data Science / AI-ML / Database
- Mode: Online + Offline (Kothrud, Pune)
- Batch Timings: Weekday and Weekend Batches — morning, afternoon, evening
- Language: English (Marathi and Hindi support available)
- Certification: Track-specific certificate at each completed level
- Eligibility: Currently enrolled in engineering, BCA, or BSc CS

Who This Is For

- Enrolled in a BE or BTech program (Computer Engineering, IT, ENTC, Mechanical, or any branch)
- Pursuing BCA or BSc Computer Science
- In any year — first year through final year — of their undergraduate program
- Looking to build practical coding skills beyond what college teaches
- Preparing for internships, on-campus or off-campus placements
- Wanting to build a strong technical portfolio, LinkedIn profile, and GitHub presence
- Interested in specialising in a particular technology track — frontend, backend, data, AI, or full stack

When This Programme Is The Wrong Choice

- Final-year students with placements already underway — there is no longer time for a semester-paced programme. TechReady is built for that situation.
- Graduates who have already finished their degree. CareerCode is structured around semesters you are actually sitting; without them it loses its shape.
- Anyone wanting a job in three months. This programme trades speed for depth deliberately.
- Students who cannot commit across terms. Taking Sem 1-2 and stopping leaves you with foundations and no specialisation.
- Students whose college workload is already unmanageable. One or two technologies a semester is light, but it is not nothing — talk to us before enrolling.

Why Join

Your Degree Alone Is Not Enough

A BE or BTech degree from most colleges in India no longer guarantees a software job. Companies look for students who can code, communicate, contribute to real projects, and hit the ground running on Day 1. Most colleges do not produce this. CareerCode does.

Technology Moves Faster Than Curriculum

Engineering syllabuses are reviewed every few years. Technology changes every few months. CareerCode is updated monthly — so you always learn what is actually in demand right now, not what was relevant five years ago.

The Internship Gap Is Real

Most engineering students reach third year without having done a single real internship or built a single deployable project. CareerCode starts building your internship readiness from Year 1.

Placements Go to the Prepared

The students who get shortlisted are those with strong coding skills, a confident interview presence, a well-structured resume, and an active professional profile. CareerCode builds all of this systematically.

Soft Skills Are a Hard Requirement

Every major company tests communication skills, logical reasoning, and aptitude in their hiring process. CareerCode integrates communication, aptitude, and mock interview training throughout.

Curriculum

Frontend Developer

Build the interfaces users see and interact with.

Sem 1-2

HTML5, CSS3, JavaScript ES6+ — the complete web foundation

Sem 3-4

React.js — components, hooks, state management, React Router

Sem 5-6

TypeScript, Tailwind CSS, REST API integration, responsive UI

Sem 7-8

Next.js, performance optimisation, SEO for web apps, deployment

Internship Prep

Portfolio projects, GitHub profile, live deployment, interview prep

Roles this track leads to:

- Frontend Developer
- UI Developer
- React Developer
- Web Designer-Developer

Backend Developer

Build the logic, databases, and systems that power applications.

Sem 1-2

Python or Java fundamentals, OOP, problem-solving patterns

Sem 3-4

REST API development with FastAPI or Spring Boot

Sem 5-6

Databases — MySQL, PostgreSQL, MongoDB; database design

Sem 7-8

Authentication, caching with Redis, Docker basics, cloud intro

Internship Prep

API project, Postman, system design basics, backend interview prep

Roles this track leads to:

- Backend Developer
- API Developer
- Java Developer
- Python Developer

Full Stack Developer

Master both frontend and backend — the most in-demand engineering profile.

Sem 1-2

HTML, CSS, JavaScript + Python or Java fundamentals

Sem 3-4

React.js (frontend) + FastAPI or Node.js (backend)

Sem 5-6

Database design + full stack integration + API consumption

Sem 7-8

Cloud deployment, CI/CD pipelines, Docker, real-world project

Internship Prep

End-to-end capstone project, GitHub, mock interviews

Roles this track leads to:

- Full Stack Developer
- Software Developer
- Application Developer

Data Science and Analytics

Work with data to extract insights and drive business decisions.

Sem 1-2

Python fundamentals, statistics, Excel for data

Sem 3-4

Pandas, NumPy, data cleaning, Matplotlib, Seaborn

Sem 5-6

Machine learning basics — Scikit-learn, model evaluation, regression, classification

Sem 7-8

SQL for analytics, Tableau, Power BI, storytelling with data

Internship Prep

Kaggle projects, portfolio datasets, dashboard projects

Roles this track leads to:

- Data Analyst
- Data Scientist
- Business Analyst
- Analytics Engineer

AI and Machine Learning Engineer

Design, build, and deploy intelligent systems.

Sem 1-2

Python, mathematics for ML — linear algebra, probability

Sem 3-4

Classical ML algorithms, Scikit-learn, model training and evaluation

Sem 5-6

Deep Learning — neural networks, TensorFlow, Keras basics

Sem 7-8

NLP, computer vision, model deployment with FastAPI and Streamlit

Internship Prep

AI mini-projects, Hugging Face, GitHub portfolio, ML interview prep

Roles this track leads to:

- ML Engineer
- AI Engineer
- Deep Learning Engineer
- NLP Engineer

Database Administrator

Design, manage, and optimise the data systems that power enterprises.

Sem 1-2

Relational databases, SQL fundamentals, ER diagrams

Sem 3-4

Advanced SQL, stored procedures, views, indexing

Sem 5-6

NoSQL databases — MongoDB, Redis; when to use which

Sem 7-8

Database administration — backups, replication, security, performance

Internship Prep

Design and document a real-world database schema project

Roles this track leads to:

- Database Administrator
- SQL Developer
- Data Engineer
- Backend Developer

Projects You Finish With

Semester coursework, done properly

The projects your syllabus already requires, built to a standard that survives a GitHub reader — version-controlled, documented, and deployed where it makes sense. Most students throw these away; treating them as portfolio pieces from first year is the single cheapest advantage available to an engineering student.

- Technology: Git
- Technology: GitHub
- Technology: READMEs
- Technology: Your track's stack

A track project at each level

Every semester block ends in something built rather than something attended — a React interface, an API with authentication, a cleaned dataset with a dashboard, a normalised schema with stored procedures. Each one is reviewed by a mentor and revised, which is the part self-study skips.

- Technology: React.js
- Technology: FastAPI or Spring Boot
- Technology: MySQL / MongoDB
- Technology: Pandas
- Technology: Power BI

Final-year capstone and internship portfolio

By Sem 7-8 you assemble the earlier work into a coherent capstone and a portfolio aimed at the roles your track targets. Because the components already exist and have been reviewed, the final year is spent refining and interviewing rather than building from zero while placements run.

- Technology: End-to-end project
- Technology: Cloud deployment
- Technology: CI/CD basics
- Technology: Portfolio site

Included With Every Track

- Communication and Business English - Speaking confidently in meetings, writing professional emails, presenting technical ideas clearly, and participating in group discussions and interviews.
- Aptitude Training - Quantitative reasoning, logical reasoning, verbal ability, and data interpretation — the four pillars tested by every major IT company. Regular practice sessions and mock tests.
- Interview Preparation - Company-specific interview patterns, common coding questions, HR rounds, technical rounds, and group discussion preparation. Mock interviews with detailed feedback.
- Resume Building - Create an ATS-friendly resume that gets shortlisted. Learn what recruiters look for, what to highlight, and how to present projects and skills effectively.
- LinkedIn Profile Optimisation - Building a complete, keyword-rich LinkedIn profile — headline, summary, skills, endorsements, recommendations, and activity strategy.
- Naukri.com Profile Management - How to set up and maintain your Naukri profile, optimise for recruiter searches, and apply strategically for internships and jobs.
- GitHub Profile Management - Maintaining a consistent, professional GitHub presence — pinned repositories, well-written READMEs, regular contributions, and project documentation.

- Internship Guidance - How to find legitimate internships, apply, prepare, and perform. Students are guided towards Archer Infotech's internship network and placement connections.

Tools and Technologies

Across the six tracks, CareerCode students work with the tools Pune's IT employers in Hinjawadi, Kharadi, Magarpatta and Baner actually run. You do not learn all of these — you learn the set your chosen track needs, one or two technologies per semester.

Web and Frontend

- HTML5
- CSS3
- JavaScript ES6+
- React.js
- TypeScript
- Tailwind CSS
- Next.js

Backend and APIs

- Python
- Java
- FastAPI
- Spring Boot
- Node.js
- REST APIs
- Redis

Databases

- MySQL
- PostgreSQL
- MongoDB
- SQL
- ER modelling
- Indexing and tuning

Data and AI

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- TensorFlow
- Keras
- Hugging Face

Analytics and BI

- Excel for data
- SQL for analytics
- Tableau
- Power BI
- Streamlit

Developer Tools

- Git
- GitHub

- Docker
- Postman
- VS Code
- Kaggle

Career Outcomes

CareerCode is built around a simple observation: students who start building in first year interview very differently in final year from students who start in final year. The programme spreads the work across the degree so that by the time placements begin, the portfolio already exists.

Immediate benefits

- Coursework becomes easier — the programming, DBMS and web subjects in your syllabus are taught here first, in more depth.
- A GitHub profile that grows every semester rather than being assembled in a panic before placements.
- Eligibility for second- and third-year internships, which in Pune go to students with visible project work.
- Hackathon and college-project readiness from first year.
- A specialisation chosen deliberately, with a trainer, rather than defaulted into at the last minute.

Longer-term paths

- Frontend Developer, UI Developer or React Developer
- Backend Developer, API Developer, Java Developer or Python Developer
- Full Stack Developer, Software Developer or Application Developer
- Data Analyst, Data Scientist, Business Analyst or Analytics Engineer
- ML Engineer, AI Engineer, Deep Learning Engineer or NLP Engineer
- Database Administrator, SQL Developer or Data Engineer

Why this matters in Pune specifically

Pune's campus hiring runs on a compressed timetable — most engineering colleges see the bulk of their drives inside a single term of final year. Students who begin preparing when the calendar is announced are competing against students who have been building for three years. CareerCode exists to put you in the second group without asking you to choose between your degree and your skills: one or two technologies a semester, taught alongside your coursework, at our Kothrud campus or online.

Indicative Pune salary bands

- Junior React Developer (Pune entry, <2 years) - Rs. 3,50,000 – Rs. 6,00,000 per year (source: AmbitionBox Pune React Developer)
- Junior Python Full Stack Developer (Pune entry, <2 years) - Rs. 4,00,000 – Rs. 7,00,000 per year (source: AmbitionBox Pune Python Full Stack Developer)
- Junior Data Analyst (Pune entry, <2 years) - Rs. 3,50,000 – Rs. 6,00,000 per year (source: AmbitionBox Pune Data Analyst)
- Junior ML Engineer (Pune entry, <2 years) - Rs. 6,00,000 – Rs. 10,00,000 per year (source: AmbitionBox Pune ML Engineer)

Companies hiring for these skills in Pune

- Persistent Systems
- Capgemini
- LTIMindtree
- Tech Mahindra
- Amdocs
- Cybage
- Cognizant
- Wipro
- Accenture
- TCS

Fees

CareerCode is billed per semester block rather than as one upfront programme, because that is how it is taken — you commit a term at a time rather than signing for the whole degree up front. Ask admissions what applies if you need to skip a block. The per-block fee depends on the track and the technologies in that block, so we quote it against your specific track and year rather than publishing a single number that would be wrong for most students.

- Per-semester-block payment rather than one upfront sum
- Instalment options within a block — ask admissions what applies
- Free demo class and a track-selection counselling session before you commit

Placement Support

CareerCode is not a placement programme and does not carry a placement guarantee. Your placements will run through your college. What this programme changes is what you bring to them: three or four years of reviewed project work, a chosen specialisation, and interview practice that started long before the drive calendar was announced.

The internship-preparation block is where this becomes concrete. Pune internships in second and third year go overwhelmingly to students with visible project work, and that is the gap CareerCode is designed to close. Students who want full placement support after graduating move to TechReady, and CareerCode graduates enter it well ahead of the intake.

The process

- Sem 3-4: first reviewed track project committed to GitHub with documentation
- Sem 5-6: aptitude and communication training begins alongside the technical track
- Sem 5-6: internship applications supported — CV, profile and project narrative
- Sem 7-8: capstone assembled, mock interviews run for your track specifically
- Sem 7-8: Naukri and LinkedIn profiles built, college placement drives prepared for

Partner companies

- Persistent Systems
- Capgemini
- LTIMindtree
- Tech Mahindra
- Amdocs
- Cybage
- Cognizant
- Wipro

CareerCode or waiting until final year — does starting early actually matter?

The honest case for waiting: you do not yet know what you want to specialise in, technologies change across four years, and final-year intensive programmes exist precisely because they work. Plenty of students are placed well having started late, and we run TechReady for exactly that path.

The case for starting early is not that the technologies stay current — some will not. It is that the habits and the artefacts do. A student who has been committing code since first year has a GitHub history, has debugged their own work hundreds of times, and has already been through choosing and abandoning a direction. In a final-year interview that student is visibly different from one who compressed the same learning into four months, and the difference shows up most in the questions that go past what was memorised. If you are in first or second year, this is the cheaper path in every sense; if you are already in final year, take TechReady instead.

Prerequisites and How to Begin

You need to be currently enrolled in an engineering, BCA or BSc CS programme — any year, any branch. No prior programming beyond whatever your semester has covered is assumed; the Sem 1-2 block starts from foundations

regardless of when you join.

The one decision worth taking seriously before enrolling is which of the six tracks to take. We run a counselling session for this rather than letting students pick by name, because the tracks lead to genuinely different jobs and switching after two semesters costs time. Come with a rough sense of whether you prefer building interfaces, building systems, or working with data.

Suggested first steps

- Book the free track-selection counselling session before choosing a specialisation
- Check your semester timetable against the batch timings — weekday and weekend options exist
- Create a free GitHub account; it becomes your portfolio from the first block
- Bring your college syllabus so we can map what overlaps and what does not
- Attend one demo class in the track you are leaning towards

How We Compare

The comparison is deliberately anonymous — these are patterns we verified across Pune institutes and the national online programmes aimed at the same students, not accusations about any named provider.

- At CareerCode: Starts in first year and runs semester by semester alongside your degree | Most other options: Typically a three-to-six-month crash course taken in final year, once placement season has already begun
- At CareerCode: One or two technologies per semester, built to sit around your college timetable | Most other options: Full-time day batches or weekend batches that compete with college for the same hours
- At CareerCode: Classroom in Kothrud, with the same trainers following you across semesters | Most other options: The largest programme aimed at first- and second-year students is online-only, with no classroom in Maharashtra
- At CareerCode: Billed per semester block, so you commit a term at a time | Most other options: The national online programmes built for this audience run roughly Rs. 1.15–3 lakh, typically financed over 24–36 months
- At CareerCode: Track chosen in a counselling session against the job it leads to | Most other options: Track chosen from a course list at sign-up, before anyone has asked what you want to do

Frequently Asked Questions

What makes CareerCode different from other programming courses?

CareerCode is not a one-time course. It is structured to run alongside your entire engineering career — semester by semester — so you are never cramming years of skills into a few weeks before placement season. It combines technical training with communication, aptitude, LinkedIn, resume, and GitHub — making it the most holistic career development program available to engineering students in Pune.

Can I join CareerCode from my first year of engineering?

Yes, and we highly recommend it. Students who start in the first year have the greatest advantage — they have time to build skills gradually, do projects, get internships, and arrive at placement season fully prepared.

Can I join CareerCode from my third or final year?

Absolutely. CareerCode has structured entry points for students at every stage. For third and final year students, we offer accelerated tracks that prioritise internship and placement readiness.

How many hours per week does CareerCode require?

CareerCode is designed to fit around your college schedule. Classes are typically held on weekends or weekday evenings. The expected time commitment is 8 to 12 hours per week, including classes, assignments, and projects.

Is online training available for CareerCode?

Yes. CareerCode is available in both classroom mode at Kothrud, Pune, and online through Archer Infotech's LMS platform — with recorded sessions for students who cannot attend live classes.

Will CareerCode help me get internships?

Yes. Internship preparation is a core part of CareerCode. We guide you through building the right profile, applying to the right opportunities, and preparing for internship interviews. Archer Infotech also has direct connections with 100+ companies for internship and job placements.

Next Step

Start your CareerCode journey today.

Four years of engineering will pass faster than you think. Every semester you wait is a semester your competition is using. The students who land the best internships and the best placements in their final year are not the ones who started preparing at the end — they are the ones who started at the beginning.

Enquire or book a free demo session. Archer Infotech, Flat No. 12, Divyadarshan Housing Society, behind Kothrud Bus Stand Road, Kothrud, Pune 411038. Phone +91 9850 678451. Email info@archerinfotech.in. Web archerinfotech.in.